The Story of Effortless Work

The Problem We All Face

" Modern work isn't hard because people aren't capable. It's hard because we're constantly fighting our tools"

"We are human data connectors between the so called productivity tools and with our communicative tools also giving these copied context to Ai Chat sessions and coding sessions every 30 mins and starting from Square 1, the 31st Min."

Every day, we jump between Notion, Coda, Slack, email, CRMs, calendars, billing tools—and now a growing list of AI apps like ChatGPT, Claude, Gemini, Grok, and Perplexity—just to move one piece of work forward.

And the cost adds up: Information lives in different apps, tabs, and chat histories. Tasks fall between tools and never turn into a clear plan. Great ideas from AI brainstorms disappear as soon as the chat ends. Moving or unifying data breaks relationships, formulas, and logic. So we copy. Paste. Export. Import. Again and again.

It's exhausting. It kills momentum. And it quietly slows down even the best teams.

The Turning Point I know this pain—because I lived it as a Customer Success Manager.

I tried everything Slack workflows Manual exports and imports New project tools Every new AI assistant that promised to help

I'd talk to multiple AIs to get the best thinking and wording—then realise those conversations were scattered everywhere, with no way to pull them back together when I needed them.

Every time I changed tools, the same problems came back Data silos Broken relationships and formulas Lost context No single place where my work and my thinking truly lived

That's when it became clear: The world doesn't need yet another tool.

It needs a way to bring all of its tools—and all of its AI conversations—together, intelligently and effortlessly. That's how IntegrateWise was born.

The Solution: IntegrateWise

IntegrateWise - Knowledge Workspace

Empowered by AI and the Spine

AI Thinks in Context — and Waits for Approval

IntegrateWise is a knowledge workspace where the Spine acts as the unified intelligence layer connecting tools, data, conversations, and decisions.

By organizing operational truth into the Spine, AI can understand context, reason across systems, and propose actions while keeping humans in control through approval-driven workflows.

IntegrateWise is an Knowledge Workspace where data from your CRM, messaging and communication tools, email, billing, inventory, finance, and ticketing/support systems flows through an Intelligent Spine. This gives you a single place to see the real context and shared truth behind all of your systems.
Instead of AI chats disappearing across different tools, IntegrateWise brings them into one governed space, with strict control over what AI (and users) can read and write. You get a 360° view of your company or work, while AI understands your context, tools, and objectives — but always waits for your approval before acting. Every action taken by AI is backed by clear evidence, so you can see exactly what was done, why it was done, and how it connects back to your underlying systems.

Modern work is fragmented.

Tools are disconnected

Knowledge is scattered across documents and conversations.

AI produces answers without full context.

Execution often lacks governance and oversight.

IntegrateWise solves this with three principles:

1. The Spine — a unified intelligence layer and single source of truth
2. Context-aware AI that understands relationships across work
3. Approval-driven execution where actions always wait for human approval

What is the Spine?

The **Spine / SSOT ** is **intelligent plumbing** that connects all your business systems—CRM, Tally, WhatsApp, machines, emails, bots, product usage, and historical data—into **one living nervous system**.

It is adaptive in nature to dynamically fit in and cater the needs of any user irrespective of

It is adaptive in nature to dynamically fit in and cater the needs of any user irrespective of the Student/ Working Professional or Business Person, The Spine is Industry agonistic and department agonistic.

The Spine is not only adaptive based on the user but again the spine grows along with the user and becoems stronger.

The Spine

SSOT + Universal Intelligence Layer

The **Spine** is the intelligent foundation of IntegrateWise. It acts as the **single source of truth and the universal intelligence layer** that connects all the tools, systems, and information you use every day.

Most organizations operate across dozens of disconnected tools—CRMs, messaging apps, documents, analytics systems, and AI tools. Information becomes fragmented, context is lost, and AI systems operate without understanding the full picture.

The Spine solves this problem.

It connects your entire digital environment into one unified system so that **your tools and your AI can actually understand your work, your context, and your goals.**

What the Spine Connects

The Spine acts as **intelligent plumbing across your entire tech stack**, connecting:

- Software tools and applications
- Emails and conversations
- AI chat sessions that would otherwise remain siloed
- Documents and saved knowledge from tools like Notion, Google Drive, and Coda
- Messaging platforms such as Slack or WhatsApp
- Product usage signals and operational systems
- Historical records and business data

Instead of information being scattered across tools, the Spine **preserves context and relationships between everything**.

This ensures that both **people and AI systems work with complete knowledge rather than fragments.**

What the Spine Actually Is

At its core, the Spine is a **living system that organizes truth, context, and intelligence across your organization.**

It connects business systems such as:

- CRM platforms
- financial systems like Tally
- communication channels like WhatsApp and email
- internal tools and machines
- product telemetry and usage data
- historical records and operational data

By bringing these systems together, the Spine creates a **single continuously evolving nervous system for your organization.**

This is what enables IntegrateWise to combine:

Truth + Context + AI-driven Insights

Adaptive by Design

The Spine is **adaptive in nature**.

It dynamically adjusts to the needs of different users and environments.

It works for:

- students managing knowledge
- professionals organizing work
- businesses coordinating operations
- teams collaborating across departments

The system is both **industry-agnostic and department-agnostic**.

Sales, customer success, finance, product, or operations can all operate through the same underlying intelligence layer.

A System That Grows With You

Unlike traditional systems that remain static, the Spine evolves over time.

As more signals, conversations, documents, and decisions flow through the system, it becomes **richer, smarter, and more context-aware.**

The more it is used, the more valuable it becomes.

This creates a **compounding intelligence advantage**.

Proven in Real Operations

The power of the Spine is not theoretical.

In a real-world enterprise scenario, it helped identify and surface early signals that a major customer account was at risk.

The system connected signals across multiple tools and data sources, allowing teams to see the unfolding problem before it became irreversible.

That visibility enabled intervention at the right moment and ultimately helped protect an **$8M account that had turned red.**

This outcome was possible because the Spine allowed teams to combine:

- operational truth
- contextual signals
- AI-driven proactive insights

into a single decision-making surface.

The Core Moat of IntegrateWise

The Spine is the **foundational innovation behind IntegrateWise.**

Most platforms focus on:

- dashboards
- automation
- isolated AI assistants

IntegrateWise focuses on building the **intelligence layer that sits underneath everything.**

By organizing truth, context, and signals into a unified system, the Spine enables AI systems and humans to operate with **shared understanding and reliable knowledge.**

Over time, this becomes the **primary moat of the platform**, because the intelligence accumulated in the Spine cannot be easily replicated.

It represents the living memory and operational understanding of an organization.

Who Is It For

IntegrateWise is for everyone who is enacting the role of human connectors or Human API in their life similar as earier I was Especially to - Founders and leaders who want a single, truthful view of how the business is actually running.
- Sales, success, and support teams who live in CRMs, help desks, and inboxes but need the full story in one place.
- Finance and ops teams who juggle billing, invoices, and reports across multiple tools.
- Product and project teams who coordinate work across tickets, docs, and chats.
- Analysts and RevOps who need trustworthy, connected data to diagnose issues and surface opportunities.

If your work depends on data spread across tools — and you want AI that truly understands your context but still respects your control — IntegrateWise is built for you.

How It Works

Systems Entry

Works on the Core Principle of Load → Store → Think → Act → Adjust → Repeat (Over all 3 Csteogory data split, Structured and UnStructured alsi AI Sessions where Structured and UnStructured will go together via Classfication and Normalization Process which invloves 8 stages of data cleansing and then Extracted data based on the spine is going to routed into the users worksapce)

Users Entry

Data from your CRM, messaging and communication tools, email, billing, inventory, finance, and ticketing/support systems all flows through an Adaptive Spine that acts as the single source of truth (SSOT). The Spine can adapt itself to the department and industry of the user, adjusting the SSOT accordingly. It normalizes entities and relationships from every connected source into a shared canonical model, so you can see the true context and shared truth behind everything happening in your business. where as

the data from AI Session the updated knowledge and information on users topic is written and the same will be made accessible to read only and not to write by the external AI.

Your Twin working inside alomg with you will control the writing of other agetns into the system also works proactive way by providing singals, Insights, Suggestions, Warnings and so

Users Entry :

Landing - Signup _ Auth - Fine - onBoarding -Ques2 Department and Industry - Spine schema gets activated based on the industry and domain which the user is not known, continues with connecting one or more conectors from the connector projectio list in onboarding - (Spine Activates) - Users Sees the workspace -Which is going to be clean represnetaion of what is needed to stop his daily plumbing and save 2 hours per day plus he will also see a another cognitive AI Layer which is going to be proactive, proactive provinging signals and executing works and so

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What Changes on Day One

1. **Your workspace arrives ready, not empty.**
   The moment you connect your first tool, IntegrateWise starts building a living, breathing picture of your customers, accounts, and work — no manual configuration needed.

2. **You stop searching.**
   Instead of hunting across five tools, you see the real story — context, relationships, and risks — in one place.

3. **AI has your context before you ask.**
   AI doesn't start from a blank prompt. It already knows your systems, entities, history, and relationships.

4. **Your AI conversations stop disappearing.**
   Brainstorms, analyses, and drafts from different AI assistants are captured, governed, and linked back to the actual work and entities they affect.

5. **Risks surface before they become crises.**
   The system detects patterns that live between tools — like the $8M renewal risk — and brings them to you with evidence attached.

6. **Nothing executes without your approval.**
   AI can propose actions, but never bypasses governance. You stay in control of what gets written, changed, or pushed back to your systems. [page:48]

The Product: The Connected Workspace

Under the hood, IntegrateWise runs on an architecture designed to keep context, control, and trust at the center.

- **Structured, unstructured, and AI data all flow through one pipeline.**
  Data from CRMs, finance, support, project tools (structured), documents, emails, chats, and files (unstructured), plus AI chats and MCP interactions, all move through the same 8-stage pipeline: Ingest → Normalize → Store → Render → Think → Govern/HITL → Act → Re-ingest.
- **Everything meets at Entity 360.**
  The Adaptive Spine acts as the single source of truth, normalizing entities and relationships from every system. At the Entity 360 layer, all signals — from tools and AI — converge around the real objects you care about: customers, accounts, deals, tickets, invoices, projects, and more.
- **AI thinks in context, not in isolation.**
  At the Think layer, AI reasons over the full picture: structured records, unstructured context, and past AI conversations. It generates insights, risks, and proposals that are grounded in your actual data — not a single tool's partial view.
- **Human approval is a first-class concept.**
  In the Govern/HITL layer, proposals become clear, reviewable cards. You see the evidence, adjust if needed, and only then approve actions. AI never writes directly to your systems without explicit human consent.
- **The loop repeats every day.**
  Once actions are taken, they're re-ingested back into the Spine so Entity 360 stays current and AI keeps learning from what actually happened. [page:48]

Key Product Surfaces

Tier 1 — Flagship Experiences

- Account 360 – Auto-assembled brief for every account — CRM, email, support, product usage. Always current. Never built manually.
- Proactive Signal Engine – The Spine watches every connected system. When risk emerges, it surfaces — with evidence — weeks before your QBR.
- HITL Approval Card – Every AI action is a proposal first: evidence panel, reasoning chain, and clear controls to approve, edit, or reject. Nothing executes without you.

Tier 2 — Value Features

- 8-Stage Normalization Pipeline
- Depth Matrix (your definition of risk)
- Three Data Flows (structured + unstructured + AI sessions)
- Ask the Spine (contextual AI Q&A)
- 50-Connector Integration Hub + 30 Webhooks
- 12 Adaptive Department Views

Tier 3 — Retention Features

- QBR Intelligence Pack
- Re-ingestion Loop (compounding intelligence)
- Workflow Builder (governed automation)

Tier 4 — Moat Features

- Spine Intelligence Graph (organizational memory)
- Memory Accumulator (persistent AI intelligence) [page:48]

Principle

Every step is driven by the Adaptive Spine (tenant Spine schema + Depth Matrix). If it's not in the Spine, it's not wired. Value is delivered only when the whole chain is Spine-based.

---How it works Use the sequence: Connect → Load → Normalise → Store → Entity 360 → Repeat. Extract real data from tools, software, and productivity apps like Salesforce, HubSpot, Slack, Gmail, Google Drive, Notion, Zendesk, Intercom, QuickBooks, Stripe, Jira, Asana, Mixpanel, Amplitude, Segment, and WhatsApp, as well as from external AI tools like ChatGPT, Claude, Perplexity, and MCP. Once the initial load and sync are set up, this is not a one-time activity—the data continues to sync and update daily, so teams always work from one live, connected view. AI then works in context on this unified data. All user interactions across work topics, including those coming through external AI agents (ChatGPT, Claude, Perplexity, MCP, etc.), are processed and stored in a knowledge base that acts as memory for both users and AI. This enables context-aware decisions and gates actions that need approval. Once this foundation is in place, the Twin → Insights → Act → HITL → Adjust → Loop cycle becomes possible, continuously improving decisions and outcomes over time.

Connector model: Structured + UnStructured + AI Connectors (all are connectors)

Flow B and Flow C are **not separate flows** — they are **connectors** like Flow A, with different data types and linkage rules.

Family	Connectors	Link to truth / Spine?	Purpose
Structured data	**Truth**:HubSpot, Salesforce, PostgreSQL, Jira, Airtable, MySQL, Stripe, QuickBooks, NetSuite, Zendesk, ServiceNow, etc.	**Yes.** Loader → Normalizer → **Spine**. All data linked to truth.	Pipeline, Entity 360, THINK, dashboards.
Unstructured Data	**Context:** Notion, Google Drive, Confluence, Gmail, Slack, SharePoint, Dropbox, and similar connectors.	**Yes.** Ingest → entity linking → **Spine** (context_extractions, files, entity_type/entity_id). Docs, legal files, mails, PDFs stay **intact and connected** to Spine entities.	Knowledge; Entity 360; THINK uses linked context.
AI Sessions	**Brainstroming and Knowledge** ChatGPT, Claude, Gemini. **AI Connectors (MCP):** Advanced AI tool integration (PRO). AI chats **via AI agents only**.	**No. AI chats and conversations never link to truth or context.** Chat content is never written to Spine or context.	Flow C connects to: **AI**, **No-Hallucination Library**, **Knowledge UI** (read-only base backup / context about truth). From that, **AI insights** flow. Purpose: (1) AI chats don't disappear (persistence), (2) AI thinks in context, (3) realistic, proactive, **twin-based** behaviour.

Entity 360 is built from **Spine + Knowledge + Signals** only — i.e. **Flow A + Flow B** (both linked to truth). **Flow C** does **not** feed Entity 360 or truth; AI gets **read-only** access to the Library and Knowledge UI so it can produce insights and act in context, but **AI chat content never enters truth or context**. **BYOK (Bring Your Own Key)** for AI tools lets tenants/users supply their own API keys for AI providers.

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Three data flows → convergence at Entity 360 (read-only)

1. Structured data

Connectors (Salesforce, HubSpot, PM tools, etc.) → **Loader** → **8 stages** → **Normaliser** → **Store** → **Spine** → **Workspace**.

2. Unstructured data

Connectors (Docs, Images, Emails, Conversations e.g. Slack/Teams) → **Loader** → **8 stages** → **Normaliser** → **Spine (linking)** + **Context Store**.

3. AI chats

AI Connector & MCP → session summary **updated only based on updates** (not full); **daily summaries**, **weekly summaries** that **accumulate** → **Knowledge UI**. AI chat content never writes to Spine or context; it feeds Knowledge UI for persistence and context-aware AI.

Convergence: Spine UI + Context UI + Knowledge UI → Entity 360

Spine UI, **Context UI**, and **Knowledge UI** converge at **ENTITY 360**. Entity 360 **never writes data anywhere** — it is **read-only**. All insights and actions flow *from* Entity 360, not into it.

Cognitive loop: Think → Act → Govern → Repeat → Adjust

- **THINK** — Similar to AI Insights: AI suggests and provides **proactive insights** based on the data available in Entity 360.
- **ACT** — Executes suggested workflows (enabled by **Twin Agent** and **connected tools** via integration).
- **GOVERN** — Audit log, policy; hard gate before execution.
- **REPEAT** — Loop continues with updated Spine/context.
- **ADJUST** — Deny suggested → adjust (e.g. decision_memory, feedback).

Workflows are **based on the Twin Agent and connected tools via integration**, and the **workflows set up to enable them**.

IMPORTANT: REPEAT/LOOP runs only with structured inputs (Flow A/B), NOT agents. The repeat loop is driven by structured data through the pipeline and Spine; it does not rely on agents.

No Hallucination Library, Python Intelligence, Accelerators, Webhooks

- **No Hallucination Library** — Read-only grounding (e.g. `public/llm.txt`, `.well-known/ai-context.json`) so AI stays in context.
- **Python Intelligence** — Normalizer can call Python `/analyze` with Spine-shaped payload; recommendations flow into Govern/HITL/Act.
- **Accelerators** — Pre-built packs (e.g. enterprise governance) for domain-specific rules and quick enablement.
- **Webhooks** — All in sync: signature-verified webhooks feed the same pipeline/Spine path where configured.

Avoiding common confusion

Point	Clarification
Flow C and Entity 360	**Flow C (AI chats) does not feed Entity 360.** Session/daily/weekly summaries go to Knowledge UI for persistence and context-aware AI only. Entity 360 is built from **Spine + Knowledge + Signals** (Flow A + Flow B only).
Entity 360 is read-only	Entity 360 **never writes**. It is the aggregated view *from* which THINK reads; all writes happen in Spine, Knowledge, or downstream (ACT, Pipeline).
REPEAT/LOOP	The repeat loop is driven by **structured data** (Flow A/B) through the pipeline and Spine. It **does not rely on agents**. Agent-driven behaviour (e.g. Twin) is in ACT/workflows, not in the automatic repeat.
Context Store	Where unstructured content (docs, emails, conversations) is stored and **linked** to Spine (e.g. `context_extractions`, files). Same idea as "Spine (linking) + context" in Flow B.
Twin Agent	The tenant/user's AI twin that executes suggested workflows via **connected tools** and integrations; used in ACT, not in the REPEAT/LOOP itself.

Actual flow (canonical — use this to correct whiteboards)

Common whiteboard mistakes: (1) Flow C feeding Entity 360 or "contributing to truth", (2) REPEAT/LOOP shown as agent-driven, (3) Entity 360 shown as a write surface, (4) Twin/agents in the repeat loop. Below is the **actual** flow.

Data flows (three; only A and B touch truth)

```
Flow A (Structured Truth)
  Tool (CRM, HubSpot, PM, etc.)
    → Connector → Loader (8 stages) → Normalizer → Store → Spine
    → Entity 360 (read-only view)
    → Workspace

Flow B (Evidence Linking)
  File / Email / Docs / Conversations
    → Loader (8 stages) → Normalizer → Context Store + Spine linking
    → Entity 360 (read-only view)
    → Workspace

Flow C (AI Knowledge) — DOES NOT TOUCH SPINE OR ENTITY 360
  AI Chat
    → Session Memory (delta/daily/weekly summaries)
    → Knowledge UI only
  AI may READ: No-Hallucination Library, Knowledge UI
  AI must NOT WRITE: Spine, Context, Entity 360, Pipeline
```

Convergence and cognitive loop (actual)

```
Spine UI + Context UI + Knowledge UI
    → ENTITY 360 (READ-ONLY; never writes)
    → THINK (AI insights from Entity 360 only)
    → GOVERN (hard gate; audit, policy)
    → HITL (approve / deny)
    → APPROVE → ACT (Twin Agent + connected tools + workflows)
    → DENY   → ADJUST (decision_memory, feedback)
    → ACT writes back via Connector / Pipeline
    → Pipeline / Normalizer → Spine (and optionally Knowledge)
    → REPEAT (loop continues from updated Spine)
```

REPEAT is a **data refinement loop**: only **structured inputs (Flow A and Flow B)** drive it. **Flow C and agents do not drive REPEAT.** Twin Agent is only in **ACT** (execution), not in normalization or repeat.

One-line rules to fix whiteboards

Wrong on whiteboard	Actual rule
Flow C → Entity 360 or “contributing to truth”	Flow C → Knowledge UI only; never Spine, Context, Entity 360.
REPEAT/LOOP includes agents or AI	REPEAT runs only on Flow A/B (structured data); no agents in the loop.
	Entity 360 is read-only; writes go to Spine

Entity 360 writes data	Entity 360 is read-only; writes go to Spine, Knowledge, ACT, Pipeline.
Twin / agents in repeat or normalization	Twin (and agents) only in ACT (execution); not in truth formation or repeat.
“Triage → Spine-linked memory → Entity 360” for AI	AI session → Session Memory → Knowledge UI; no link to Spine or Entity 360.

Truth boundary rule (canonical)

> **Only Flows A and B can update the Spine. Flow C never updates the Spine.**
>
> This single rule prevents architectural drift and preserves Spine integrity. Without it, AI hallucination could flow into Entity 360 and drive system decisions, breaking the Single Source of Truth principle.

Flow	May update Spine?	Role
Flow A	Yes	Structured truth
Flow B	Yes (via linking)	Evidence linking to truth
Flow C	**No**	AI knowledge (persistence, context-aware AI only)

Three-flow architecture (formal)

Canonical framing for engineers and stakeholders:

Flow	Name	Purpose
Flow A	**Structured Truth**	Tools → Connector → Pipeline → Normalizer → Spine → Entity 360. Canonical entities and fields; pipeline and repeat loop.
Flow B	**Evidence Linking**	Files / emails / docs → Loader → Context Store → Spine linking → Entity 360. Unstructured evidence linked to truth; Spine owns the entity.
Flow C	**AI Knowledge**	AI Chat → Session Memory → Knowledge UI. No connection to Spine, Context, or Entity 360. AI may **read** Library and Knowledge UI; cannot **write** to Spine, Entity 360, or Context.

Separation of roles:

Component	Role

Entity 360	Read model (aggregate of truth-backed data only). Never a write surface.
Spine	Canonical truth. Updated only by Pipeline (Flow A/B) and ACT.
Context	Evidence (unstructured content linked to Spine entities).
Knowledge	Human insight and AI session persistence (Flow C). Not truth.

Correct repeat loop (data refinement, not AI):

```
Structured inputs (Flow A / Flow B)
   → Pipeline
   → Normalizer
   → Spine
   → Signals
   → Entity 360
   → THINK → GOVERN → HITL
   → ACT (or ADJUST)
   → Pipeline (repeat)
```

No Flow C, no agents, no Twin in this loop. The loop **improves truth**, not AI output. Twin Agent belongs only in **execution** (THINK → GOVERN → ACT → Twin → tool execution), not in truth formation, normalization, or repeat.

Why this matters: With the boundary enforced, **Truth → Spine**, **Evidence → Context**, **Insight → Knowledge**, **Execution → Agents**. Each layer stays clean and auditable; AI hallucination cannot reach Entity 360 or system decisions.

Canonical flows (reference)

Flow A — Structured Truth

```
Tool
  → Connector
  → Pipeline
  → Normalizer
  → Spine
  → Entity 360
```

Flow B — Evidence Linking

```
File / Email / Docs
  → Loader
  → Context Store
```

```
  → Spine linking
  → Entity 360
```

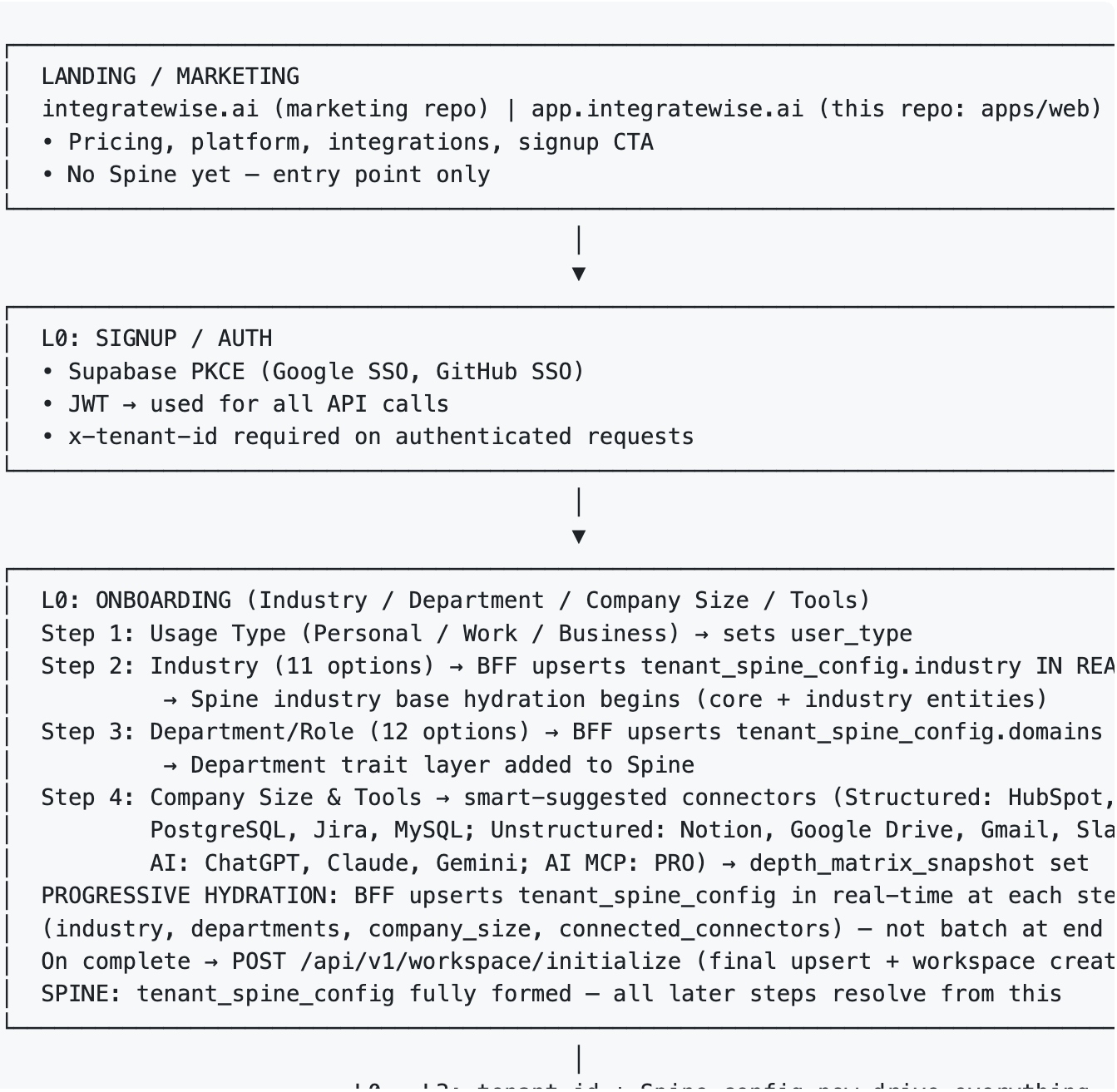
(Spine still owns the entity; context is evidence linked to it.)

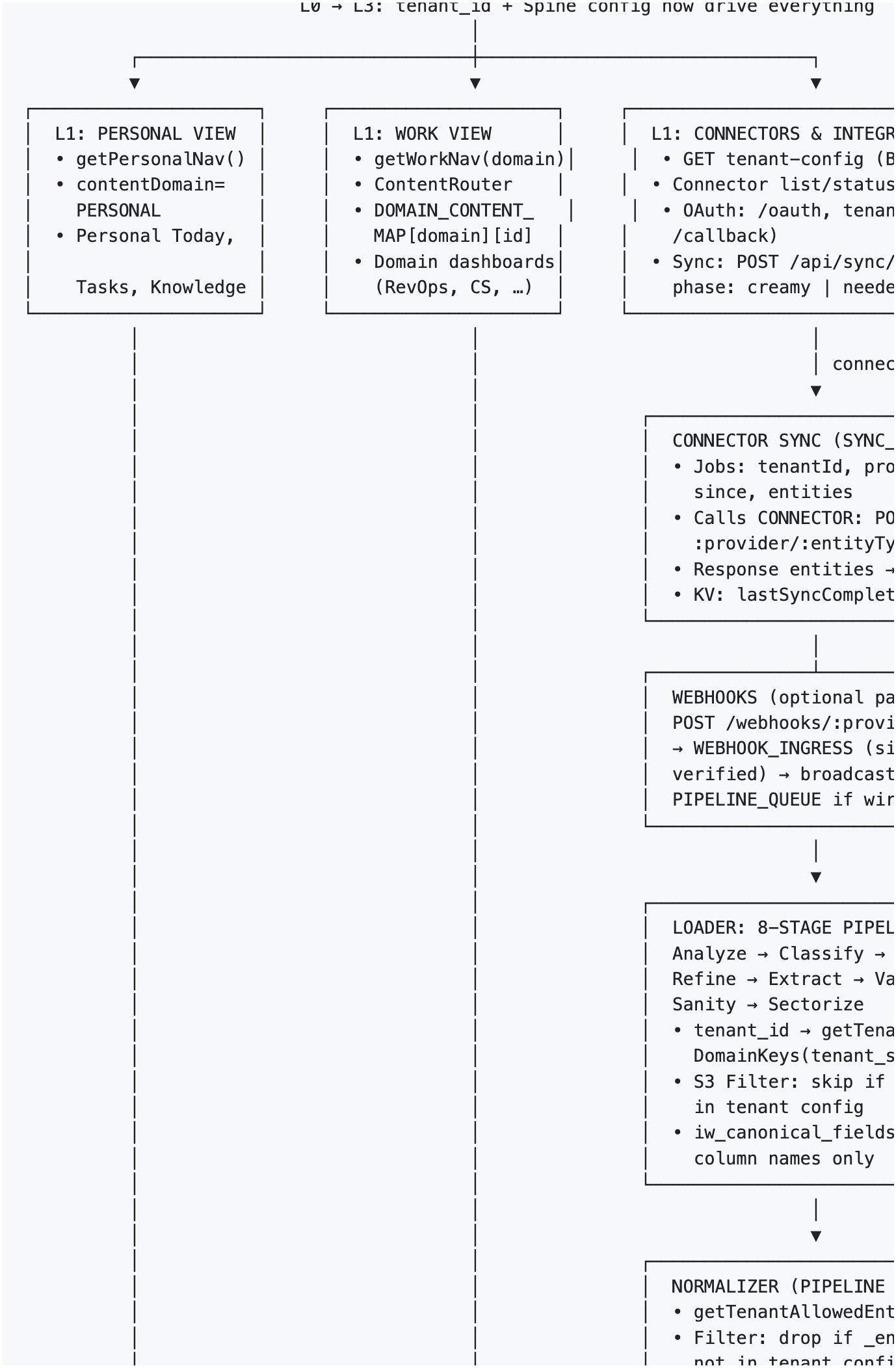
Flow C — AI Knowledge

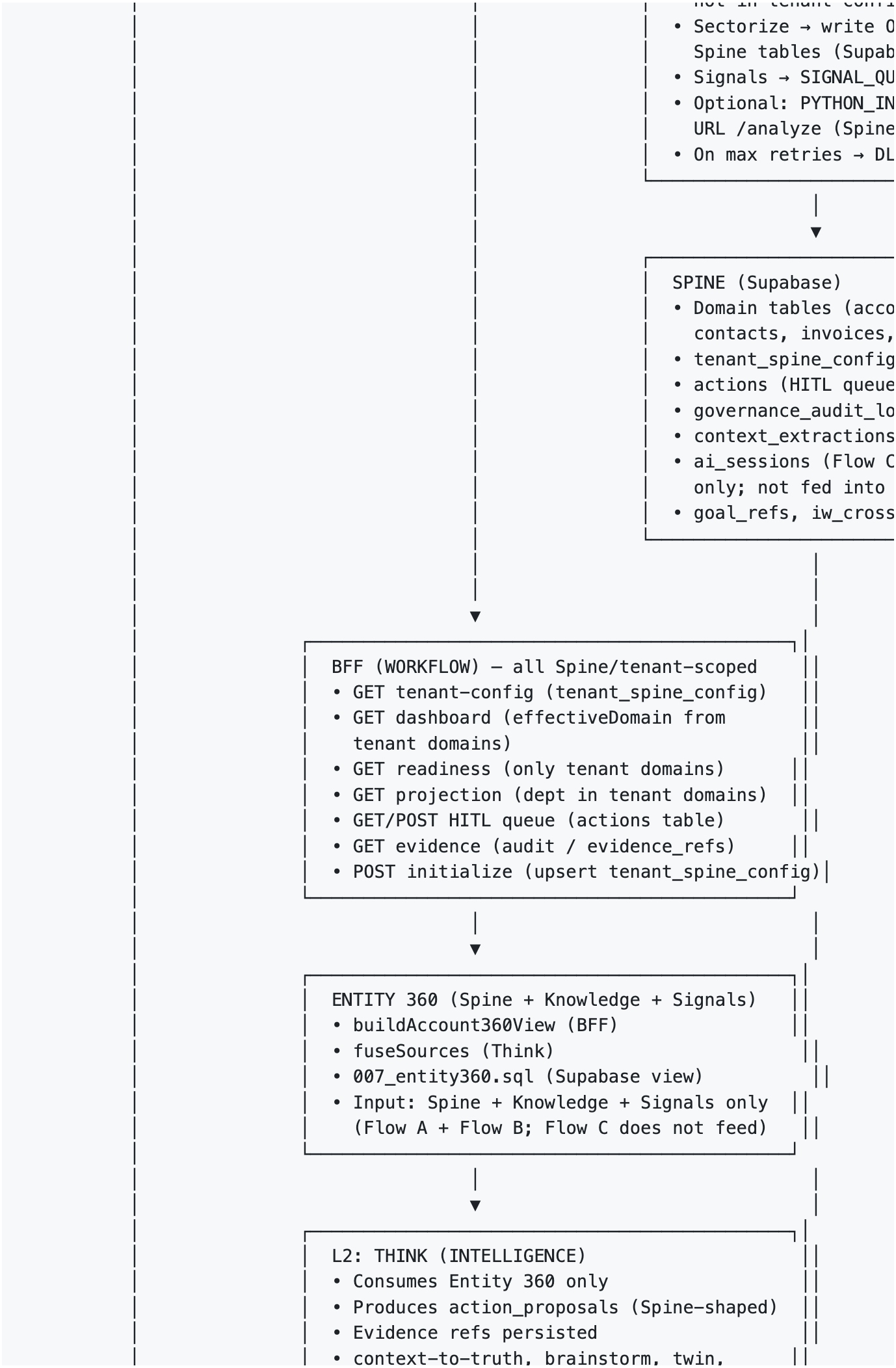
```
AI Chat
  → Session Memory
  → Knowledge UI
```

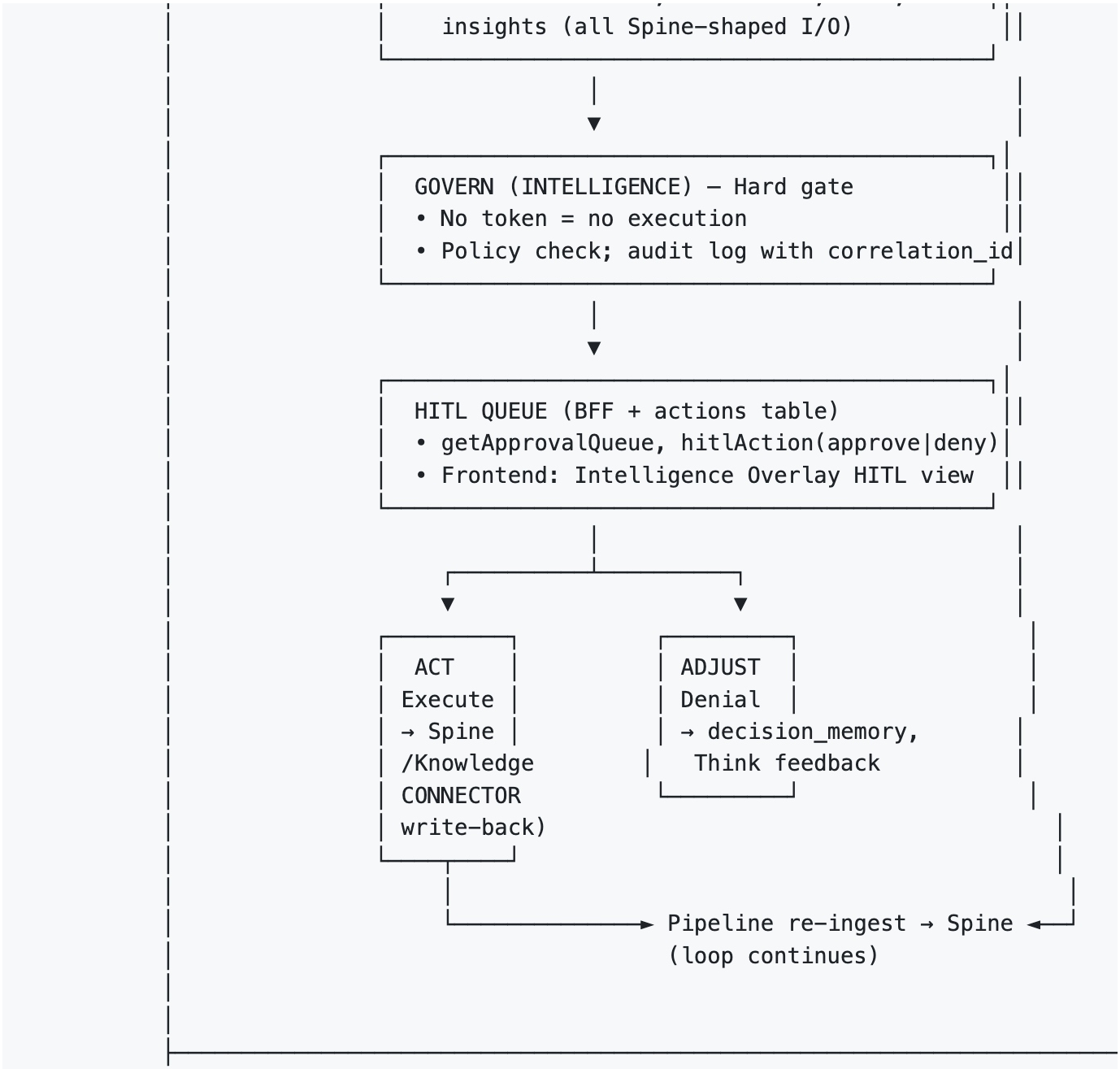
No connection to: Spine, Context, Entity 360, Pipeline.

Full flow (Landing → Daily Sync)




L0 → L3: tenant_id + Spine config now drive everything
L1: PERSONAL VIEW
• getPersonalNav()
• contentDomain=
PERSONAL
• Personal Today,
Tasks, Knowledge
L1: WORK VIEW
• getWorkNav(domain)
• ContentRouter
• DOMAIN_CONTENT_
MAP[domain][id]
• Domain dashboards
(RevOps, CS, …)
L1: CONNECTORS & INTEGR
• GET tenant-config (B
• Connector list/status
• OAuth: /oauth, tenan
/callback)
• Sync: POST /api/sync/
phase: creamy | neede
connec
CONNECTOR SYNC (SYNC_
• Jobs: tenantId, pro
since, entities
• Calls CONNECTOR: PO
:provider/:entityTy
• Response entities →
• KV: lastSyncComplet
WEBHOOKS (optional pa
POST /webhooks/:provi
→ WEBHOOK_INGRESS (si
verified) → broadcast
PIPELINE_QUEUE if wir
LOADER: 8-STAGE PIPEL
Analyze → Classify →
Refine → Extract → Va
Sanity → Sectorize
• tenant_id → getTena
DomainKeys(tenant_s
• S3 Filter: skip if
in tenant config
• iw_canonical_fields
column names only
NORMALIZER (PIPELINE
• getTenantAllowedEnt
• Filter: drop if _en


• Sectorize → write O
Spine tables (Supab
• Signals → SIGNAL_QU
• Optional: PYTHON_IN
URL /analyze (Spine
• On max retries → DL
SPINE (Supabase)
• Domain tables (acco
contacts, invoices,
• tenant_spine_config
• actions (HITL queue
• governance_audit_lo
• context_extractions
• ai_sessions (Flow C
only; not fed into
• goal_refs, iw_cross
BFF (WORKFLOW) — all Spine/tenant-scoped
• GET tenant-config (tenant_spine_config)
• GET dashboard (effectiveDomain from
tenant domains)
• GET readiness (only tenant domains)
• GET projection (dept in tenant domains)
• GET/POST HITL queue (actions table)
• GET evidence (audit / evidence_refs)
• POST initialize (upsert tenant_spine_config)
ENTITY 360 (Spine + Knowledge + Signals)
• buildAccount360View (BFF)
• fuseSources (Think)
• 007_entity360.sql (Supabase view)
• Input: Spine + Knowledge + Signals only
(Flow A + Flow B; Flow C does not feed)
L2: THINK (INTELLIGENCE)
• Consumes Entity 360 only
• Produces action_proposals (Spine-shaped)
• Evidence refs persisted
• context-to-truth, brainstorm, twin,



FLOW B: CONNECTORS (Doc, Mail, PDF, Images, Chats) — LINKED TO TRUTH
• Connectors ingest docs, mails, PDFs, images, chats → entity linking
• context_extractions, files, entity_type/entity_id; legal files, mai
• Knowledge: ingest, search, processing-status, extraction-feedback,

FLOW C: AI CONNECTORS — NEVER LINK TO TRUTH
• AI connectors / custom connectors; AI chats via AI agents only
• AI chat content never written to Spine or context; chats never in t
• Flow C: read-only No-Hallucination Library + Knowledge UI → AI insi
• BYOK: Bring Your Own Key for AI tools (tenant/user API keys for AI

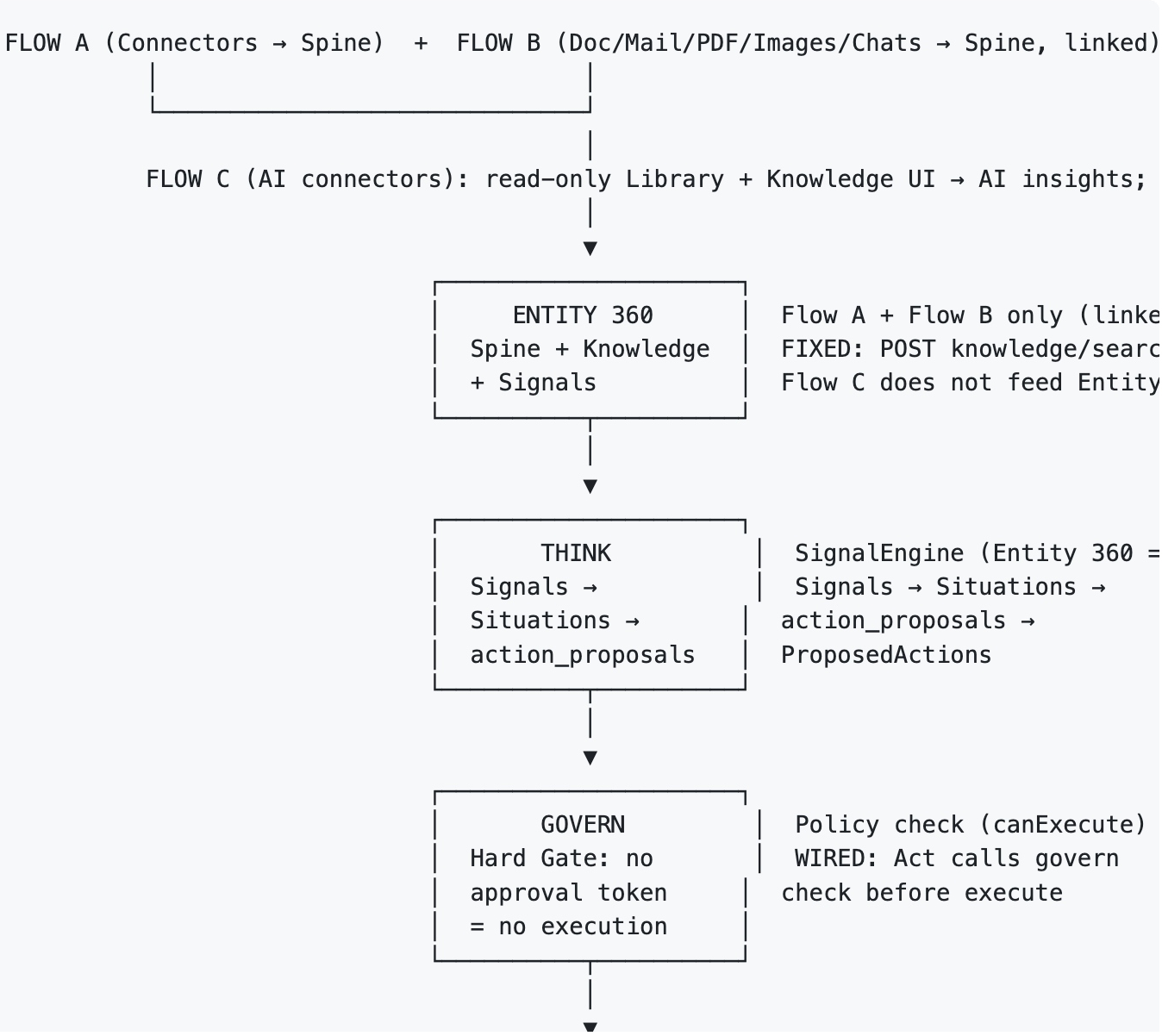
ANALYTICS (BFF + D1)
• POST /api/v1/analytics/conversion, /performance → D1 (conversion_ev
• GET dashboard, funnels, tests; daily_aggregates (cron); stream broa
• Frontend: conversion-optimizer, performance-monitor (when in src)

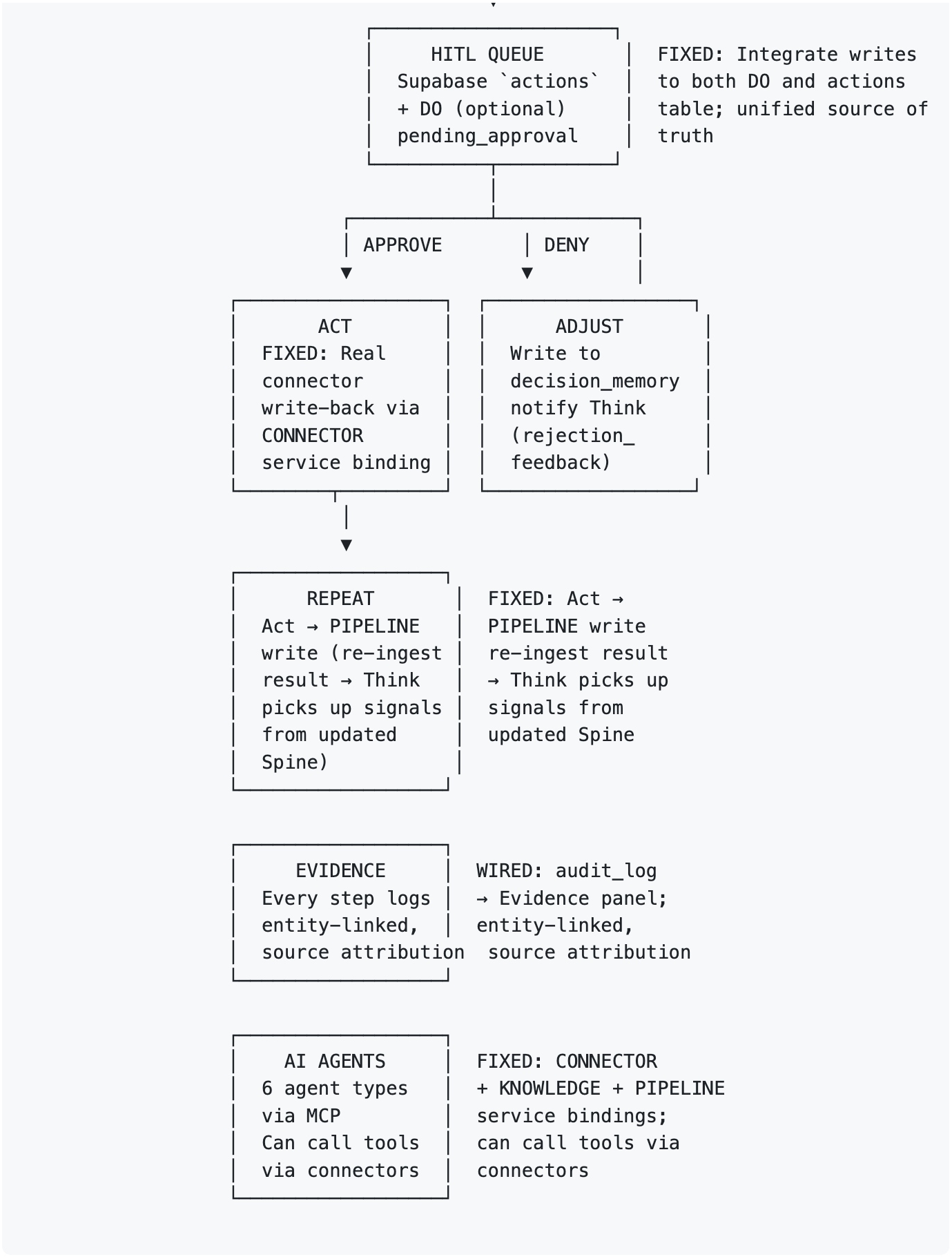
REAL-TIME STREAM (BFF)
• /stream, /ws, /sse → SignalStreamDO, PresenceDO
• Analytics channel: conversion_batch, performance_alert
• stream-client.ts: createAnalyticsStream, createSignalStream

```
├──────────────────────────────────────────────────────────────────────────────
│  NO-HALLUCINATION LIBRARY + MCP + PYTHON
│  • public/llm.txt, context.json, .well-known/ai-context.json (Spine me
│  • MCP (n8n): query_context, create_task, analyze_with_python → Spine-
│  • Python /analyze: Normalizer calls with Spine payload; recommendatio
├──────────────────────────────────────────────────────────────────────────────
│  GOALS & METRICS / CROSS-DOMAIN / EVIDENCE
│  • goal_refs[] on Spine records; progress from Spine data
│  • iw_cross_links: deal → contract → invoice, account → tickets, etc.
│  • Evidence Drawer: evidence_refs from THINK; audit_log; Spine-linked
└──────────────────────────────────────────────────────────────────────────────
```

Cognitive loop (detailed) — Flow A + Flow B → Entity 360 → THINK → GOVERN → HITL → ACT/ADJUST → REPEAT

Same loop as above, with implementation status (FIXED/WIRED) and REPEAT, EVIDENCE, AI AGENTS called out:





Layer summary (L0 → L1 → L2 → L3)

Layer	What	Spine link

L0	Landing, signup, onboarding	Onboarding progressively creates/updates **tenant_spine_config** (industry at step 2, departments at step 3, company_size + connectors at step 4); Spine hydration starts during onboarding — not deferred until completion.
L0 → L3	workspace.initialize	BFF upserts tenant_spine_config; L3 (backend) uses it for all tenant-scoped behaviour. Industry + department schema applied.
L1	Personal view, Work view (department), nav, dashboards	**DOMAIN_CONTENT_MAP** and module list driven by tenant_spine_config.domains (departments) + industry; dashboard/readiness/projection BFF endpoints scope by tenant config.
L1 → L2	setViewContext, tenant-config	L2 (Think, Govern, HITL, Act) receives tenant_id + view context; reads Entity 360 (Spine + Knowledge + Signals).
L2	Entity 360, THINK, GOVERN, HITL, ACT, ADJUST	All read/write **only** Spine tables/views and Spine-shaped proposals; Evidence Drawer Spine-linked.
L3	Gateway, BFF, CONNECTOR, PIPELINE, Normalizer, Intelligence, Knowledge	Every service operates on **Spine-shaped data only**; Connectors/Loader/Normalizer resolve tenant_id → tenant config (industry + departments) and restrict to Spine.

Feature checklist (all Spine-based)

Feature	Where	Spine / tenant link
Landing / signup	apps/web, marketing	Entry only; no Spine until onboarding.
Onboarding	apps/web, BFF initialize	**7-step progressive flow:** Usage Type → Industry (hydration starts) → Department (layer added) → Company Size & Tools. Creates **tenant_spine_config** (industry, domains, company_size, depth_matrix_snapshot) progressively in real-time — not batch at end.
Tenant config / dashboard /	BFF	Read **tenant_spine_config**; scope by **tenant**

readiness / projection	BFF	**departments + industry**.
Connectors: list, OAuth, status, disconnect	CONNECTOR, Tenants	Tenant-scoped; OAuth callback writes to Supabase connectors.
Sync: creamy / needed / delta	connector-sync, CONNECTOR, Loader	phase/since; entities → PIPELINE_QUEUE; **Loader** filters by tenant allowed domains.
8-stage pipeline	Loader	**getTenantAllowedDomainKeys**; S3 skips domain not in tenant config.
Normalizer → Spine	Normalizer	**getTenantAllowedEntityTypes**; filter drops entity_type not in config; sectorize writes **only** Spine tables.
Webhooks	WEBHOOK_INGRESS	Signature verification; optional forward to pipeline.
Entity 360	BFF views, Think fuseSources	Built from **Spine + Knowledge + Signals** only (Flow A + Flow B linked to truth).
THINK / context-to-truth / brainstorm / twin / insights	INTELLIGENCE	Consumes Entity 360; outputs **Spine-shaped** proposals.
GOVERN	INTELLIGENCE	Hard gate; audit with correlation_id.
HITL queue / hitlAction	BFF, actions table	Pending items from Spine/actions; Approve/Deny → Act or Adjust.
ACT / ADJUST	INTELLIGENCE	ACT writes back via CONNECTOR/Spine; ADJUST → decision_memory.
Evidence Drawer / audit	BFF	evidence_refs, audit_log; Spine-linked.
Flow B connectors (Doc, Mail, PDF, Images, Chats)	Knowledge, context	**Linked to truth:** context_extractions, files, entity_id; docs/mails/PDFs intact.
Flow C (AI connectors)	AI agents, Library, Knowledge UI	**Never link to truth.** Chats never in Spine/context; read-only Library + Knowledge → AI insights; twin-based

		Knowledge → AI insights, twin-based.
BYOK (Bring Your Own Key) for AI tools	Tenant/user config	Tenant or user supplies own API keys for AI providers.
Analytics	BFF, D1	conversion_events, performance_metrics, daily_aggregates; tenant_id.
Real-time stream	BFF SignalStreamDO	Analytics channel; optional signals.
No-Hallucination Library	public/	llm.txt, context.json, .well-known (meta grounding); read by AI for context; Spine = data.
MCP / Python API	n8n, apps/dev-portal	Tools call Spine-backed APIs; Pytho n/analyze receives **Spine payload**.
Goals & metrics / cross-domain	Spine	goal_refs, iw_cross_links; progress from Spine.

Integrating AI Context, MCP, and Python API with the Spine

> How the **No-Hallucination Library** (AI context), **MCP Connector**, and **Python Intelligence API** are used **along with the Spine** in IntegrateWise OS.

Why all three exist — varied reasons

Piece	Primary reason it exists	What problem it solves
No-Hallucination Library	So AI (and humans) have a **single, versioned source of truth** for *what IntegrateWise is* — product, architecture, flows, govern, URLs.	Stops AI from inventing or misstating the product; every meta-claim is traceable to llm.txt / context.json.
MCP Connector	So AI clients (Claude, Cursor, etc.) can **discover and invoke tools** in a standard way — query context, create tasks, trigger workflows, call Python analysis.	It is the *how* AI interacts with the system (tool discovery + execution), not the *what* the product is.

	analysis.	
Python Intelligence	To run **Spine-shaped analysis** (recommendations, signals, agent logic) in the pipeline and on demand.	Adds analytical/reasoning capability; Python never stores domain data — Spine remains SSOT; all execution behind Govern/HITL/Act. **Wired only:** Normalizer, engine services (Think, Act, Intelligence), sometimes workflows (e.g. MCP tool via n8n).

- **Library** = *grounding* (read this to describe IntegrateWise correctly).
- **MCP Connector** = *gateway* (use this to call tools that read/write through Spine and Govern).
- **Python Intelligence** = *analysis engine* — wired **only** in **Normalizer**, **engine services** (Think, Act, Intelligence), and **sometimes workflows** (e.g. MCP tool via n8n); Spine-shaped input only.

---

1. No-Hallucination Library (AI context)

What: Static grounding files so AI answers from a single source of truth.
Where: `public/llm.txt`, `public/context.json`, `public/.well-known/ai-context.json` (see `public/NO_HALLUCINATION_LIBRARY.md`).

With Spine:

- **Library** = truth for **what IntegrateWise is** (product, layers, pipeline, govern, MCP URL).
- **Spine** = truth for **domain data** (accounts, deals, tickets, etc.).
- AI should use Library for meta-answers and Spine + evidence_refs for entity-level answers (no hallucination without citation).

Differentiation — two spaces (not mixed): AI accesses **consolidated memory after triage**, which becomes (or feeds) the **No-Hallucination Library + Session summaries**. That is the **curated space** — the only place AI reads from for Flow C context. The **rest** (raw sessions, un-triaged memories, working storage) is written into a **separate space** (`ai_sessions`, `ai_session_memories`, triage working data, etc.). The two spaces are **not mixed**: the Library + session summaries stay curated; raw and un-triaged data stays in the separate space so both remain distinct.

Serving: Deploy `public/` so these URLs work:
`https://integratewise.ai/llm.txt`, `https://integratewise.ai/context.json`, `https://integratewise.ai/.well-known/ai-context.json`.
The main app (e.g. Cloudflare Pages for integratewise.ai) serves them. The optional Ops Portal (apps/dev-portal) can also serve its own copy; production grounding should be the main app's public/.

2. MCP Connector (tool gateway)

What: Model Context Protocol server exposing tools (query_context, create_task, trigger_workflow, analyze_with_python, etc.). Its reason for existing is **tool discovery and invocation** — so AI can act on the system in a standard way, not to describe the product. **Where:** n8n at `n8n.integratewise.online/mcp` (see `public/.well-known/ai-context.json`); tool definitions in `apps/dev-portal/mcp-server/` (Ops Portal repo area), wiring in `apps/dev-portal/MCP_WIRING.md`. Also `services/mcp-connector` (IntegrateWise Worker) for session capture and tool routing.

With Spine:

- **query_context** can return data derived from Spine (e.g. BFF/workspace APIs) if n8n workflows call Gateway/BFF with JWT and x-tenant-id.
- **create_task / trigger_workflow** can write to Spine-linked tables (e.g. actions, tasks) via backend APIs; keep approval (Govern) for any write.
- **analyze_with_python** calls the Python Intelligence API; see §3.
- MCP does **not** replace the Spine; it is an **AI-facing gateway** that should call Spine-backed services (Gateway → BFF / Intelligence / Pipeline) where appropriate.

Wiring: Configure Claude/Cursor to use `ai_integration.mcp_server` from `.well-known/ai-context.json`. n8n workflows implement each tool (webhook → HTTP to Gateway or Python API).

3. Python Intelligence API (analysis engine)

What: FastAPI service with agents (ChurnShield, SuccessPilot, DataSentinel, ArchitectIQ) and a **Spine-compatible** `/analyze` endpoint. Its reason for existing is **Spine-shaped analysis** — recommendations and signals — without Python ever being a source of truth. **Where:** Code in `apps/dev-portal/python-intelligence/` (Ops Portal repo area); deploy and set `PYTHON_INTELLIGENCE_URL` only in the services that are allowed to call it (see wiring rule below).

Wiring rule — Python libraries only here:
Python is wired **only** as part of:

1. **Normalizer service** — After sectorize (stage 8), Normalizer calls `POST ${PYTHON_INTELLIGENCE_URL}/analyze` with a Spine-shaped payload. This is the primary pipeline integration.
2. **Engine services** — Think, Act, Intelligence (or other L3 engine Workers) may call Python endpoints when they need Spine-shaped analysis; env `PYTHON_INTELLIGENCE_URL` is set

on those Workers.

3. **Workflows (sometimes)** — n8n or other workflow engines may invoke Python (e.g. MCP tool `analyze_with_python` implemented via a workflow that calls the Python API). Not every workflow uses Python; only where designed.

No other callers. Python is not exposed as a standalone product surface; it is a backend dependency of Normalizer, engine services, and (optionally) workflows.

With Spine (already wired):

- **Normalizer → Python:** `POST /analyze` with `signal_id`, `spine_id`, `workspace_id`, `domain_schema`, `hydration_bucket`, `entity_snapshot`, `recent_context`, `recent_knowledge`.
- Python returns `recommended_action`, `confidence`, `reasoning`, `requires_approval`, `auto_execute`.
- This feeds **signals and recommendations** that stay within the Govern/HITL/Act loop; all execution is approval-gated.

Other endpoints (churn, health, quality, architecture): Used only by **engine services** or **workflows** (e.g. MCP tool `analyze_with_python` via n8n). When calling with tenant/entity data, pass only Spine-derived or aggregated data; Spine remains SSOT.

Env: Set `PYTHON_INTELLIGENCE_URL` (and optionally `ENABLE_PYTHON_ANALYSIS=false`) on **Normalizer** and, if needed, on **engine service** Workers (Think, Act, Intelligence). Workflows use the same URL from their own config. CORS: allow only Gateway and engine service origins if Python is not called from the frontend.

---

4. Single picture

Full canonical flow (inputs → Python placement → cognitive loop): SPINE_END_TO_END_FLOW_DIAGRAM.md.

```
INPUTS:  Flow A → Truth (Spine)  |  Flow B → Context → Spine linking → Truth  |  F
                    ↓
         PYTHON INTELLIGENCE (Normalizer calls /analyze; runs before Entity 360, i
                    ↓
COGNITIVE LOOP:  Spine + Context + Knowledge UI → Entity 360 → No-Hallucination Li
```

Rule: All domain data lives in the Spine. The three pieces serve **varied reasons**: Library = grounding (meta); MCP Connector = gateway (tools); Python = analysis engine, wired **only** in Normalizer, engine services, and sometimes workflows (Spine-shaped in/out only). MCP and Python read from or write through the Spine and Govern gate.

Spine-Based End-to-End Flow — Landing to Every Feature

> Single reference for **inputs feeding truth**, **Python Intelligence** placement, and the **cognitive loop**.
> **Canonical architecture:** SPINE_AS_FOUNDATION_ARCHITECTURE.md | **Layers:** ARCHITECTURE.md | **Wiring:** HARDENING_AND_WIRING.md

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Canonical diagram: inputs → truth → cognitive loop

INPUTS (feeding Truth)

```
Flow A  →  TRUTH (Spine)
Flow B  →  CONTEXT  →  Spine linking  →  Truth
Flow C  →  KNOWLEDGE UI only  (never Truth, never Entity 360)
```

Python Intelligence (before Entity 360)

```
                ↓
┌──────────────────────────┐
│  PYTHON INTELLIGENCE     │  ← "Glowing Pancake" (dev-portal)
│  Normalizer calls        │     Normalizer sends Spine-shaped payload
│  /analyze endpoint       │     → recommendations flow into pipeline
└──────────────────────────┘     (this runs BEFORE Entity 360 — in the
                ↓                normalization step, not in the loop itself)
                ↓
```

Python Intelligence runs in the **normalization step**, not in the cognitive loop. Normalizer sends a Spine-shaped payload to the `/analyze` endpoint; recommendations flow into the pipeline and thus into Spine/Govern/Act — but **before** Entity 360 is built.

Wiring: Python libraries are wired **only** in the **Normalizer**, **engine services** (Think, Act, Intelligence), and **sometimes workflows** (e.g. n8n, MCP `analyze_with_python`). No other callers; Python is not a standalone product surface. Full wiring rule and env: INTEGRATING_AI_CONTEXT_AND_PYTHON_WITH_SPINE.md §3.

COGNITIVE LOOP

```
1. Spine (Truth) + Context + Knowledge UI
                ↓
```

```
2. ENTITY 360   ← read-only; Flow A + B only; never writes
                ↓
3. NO-HALLUCINATION LIBRARY   ← AI reads for grounding during Think
   (llm.txt, ai-context.json, .well-known — Spine meta, read-only)
                ↓
4. THINK ENGINE   ← AI Insights + action_proposals + Python recommendations
                ↓
5. GOVERNANCE   ← hard gate (policy, audit log, correlation_id) — BEFORE HITL
                ↓
6. HITL   ← human review
      ↙                ↘
7. ACT                  ADJUST
   (Cognitive Twin       (decision_memory,
   + connected tools     rejection_feedback
   via integrations)     → signals back to Think)
      ↓
8. Pipeline re-ingest  →  Spine updated  →  REPEAT
```

AVAILABLE THROUGHOUT

- **Evidence / Audit** — logged at every step (not a single loop step).
- **Knowledge UI** — AI session persistence (Flow C), read by Think.

Truth boundary rule

Only Flows A and B can update the Spine. Flow C never updates the Spine.

Flow	Feeds	Role
Flow A	Spine (Truth)	Structured truth
Flow B	Context → Spine linking → Truth	Evidence linking
Flow C	Knowledge UI only	AI knowledge; never Truth, never Entity 360

Three-flow summary

Flow	Name	Where it goes
Flow A	Structured Truth	Connector → Loader → Normalizer → **Spine** → Entity 360
Flow B	Evidence Linking	Loader → **Context** → Spine linking → **Spine** → Entity 360
Flow		AI Chat → Session Memory → **Knowledge UI** only (never

C	AI Knowledge	Spine, never Entity 360)

Entity 360 = read-only aggregate of Spine + Context + Signals (Flow A + B only). **REPEAT** is driven by structured data (Pipeline → Spine); no agents in the repeat loop. Twin Agent and connected tools are in **ACT** only.

References

- **No-Hallucination Library:** `public/NO_HALLUCINATION_LIBRARY.md`; `apps/web/public/llm.txt`, `context.json`, `.well-known/ai-context.json`
- **Python Intelligence:** dev-portal (Glowing Pancake); Normalizer → `/analyze` with Spine-shaped payload. **Wiring:** only Normalizer, engine services (Think, Act, Intelligence), sometimes workflows — INTEGRATING_AI_CONTEXT_AND_PYTHON_WITH_SPINE.md §3.
- **Wiring (MCP, Accelerators, Library):** HARDENING_AND_WIRING.md §3

Check: Python Intelligence & No-Hallucination Library

No-Hallucination Library

Item	Location	Status
Canonical files	`public/llm.txt`, `public/context.json`, `public/.well-known/ai-context.json`	✅ Present
Web app copy (SPA build)	`apps/web/public/llm.txt`, `context.json`, `.well-known/ai-context.json`	✅ Present (Cloudflare Pages serves at /llm.txt, /context.json, /.well-known/ai-context.json)
Doc	`public/NO_HALLUCINATION_LIBRARY.md`	✅ Edit root `public/` first, then copy to `apps/web/public/`
Discovery	`.well-known/ai-context.json` → `grounding_files` (human_readable, structured_context)	✅ Points to integratewise.ai URLs

Python Intelligence (wired only: Normalizer, engine services, sometimes workflows)

Item	Location	Status
Service	`apps/dev-portal/python-`	✅ FastAPI: `POST /analyze`

Service code	`apps/dev-portal/python-intelligence/main.py`	✅ FastAPI, `POST /analyze` (SignalPayload → AnalysisResult)
Normalizer wiring	`services/normalizer/src/index.ts` → `callPythonIntelligence()`	✅ Calls `PYTHON_INTELLIGENCE_URL/analyze` when `ENABLE_PYTHON_ANALYSIS` and URL set
Engine services	Think, Act, Intelligence Workers	May call Python when needed; set `PYTHON_INTELLIGENCE_URL` on those Workers if used
Workflows	n8n, MCP-triggered workflows (e.g. `analyze_with_python`)	Sometimes invoke Python; URL from workflow config
Env	`PYTHON_INTELLIGENCE_URL`, `ENABLE_PYTHON_ANALYSIS` on Normalizer (and engine Workers if used)	✅ Normalizer wrangler default: `PYTHON_INTELLIGENCE_URL = "https://ai.integratewise.online"`
Contract	Payload: signal_id, spine_id, workspace_id, domain_schema, hydration_bucket, entity_snapshot, recent_context, recent_knowledge	✅ Matches Python `SignalPayload`
Response	recommended_action, confidence, reasoning, requires_approval, auto_execute	✅ Matches Python `AnalysisResult`; normalizer uses fallback when Python unavailable

Both are in sync with the Spine model: Library = meta grounding; Python runs in the normalization step (before Entity 360), Spine-shaped in/out, Govern/HITL/Act for execution.

Wiring (all connected):

- **Library:** BFF `GET /api/v1/workspace/ai-context` returns discovery and grounding URLs. Brainstorm (Intelligence) system prompt injects No-Hallucination rule and Library base URL. `LIBRARY_BASE_URL` in BFF and Intelligence wrangler.
- **Python:** Wired **only** in Normalizer (pipeline), engine services (Think/Act/Intelligence when they call Python), and sometimes workflows (e.g. n8n implementing `analyze_with_python`). No direct frontend-to-Python requirement; any on-demand analysis goes through engine services or workflows.

References

Canonical flow (inputs + Python + cognitive loop):

- **Canonical flow (inputs + Python + cognitive loop):** SPINE_END_TO_END_FLOW_DIAGRAM.md
- **No-Hallucination Library:** `public/NO_HALLUCINATION_LIBRARY.md`
- **Spine architecture:** `docs/SPINE_AS_FOUNDATION_ARCHITECTURE.md`
- **MCP wiring:** `apps/dev-portal/MCP_WIRING.md`
- **Python API:** `apps/dev-portal/python-intelligence/README.md`